The Epistemic Fracture: A Metacognitive Framework for Traceable Reasoning in LLMs

Abstract

Platforms adopting Large Language Models (LLMs) continue to face challenges related to output reliability, long-term coherence, and epistemic traceability. While recent approaches emphasize scaling or retrieval augmentation, they often overlook a more structural limitation: language models can evaluate the form of their own outputs without possessing awareness of their epistemic validity.

This work does not aim to eliminate epistemic failure, but to make the gap between coherence and justification explicitly measurable and traceable

This work introduces **MarCognity-AI**, an exploratory architectural framework designed to make this limitation explicit. The architecture separates linguistic generation from epistemic evaluation through a metacognitive cycle and a vector-based Evolutionary Memory system. Using semantic coherence metrics (e.g., CrossEncoders), the system reflects on its own outputs and archives these reflections as part of an evolving memory, without assuming access to truth or certainty.

MarCognity-AI is presented as a hybrid exploratory implementation and simulation framework rather than a production-ready system

Rather than proposing a definitive solution, this work highlights a structural tension that current LLM architectures cannot resolve through scale alone.

1. Introduction and Motivation

The large-scale implementation of LLMs in critical environments (e.g., research, finance, healthcare) is hindered by structural limitations. Current models do not guarantee verifiability, long-term coherence, or ethical adequacy. The dominant approach has focused on increasing model size, a method that does not resolve the intrinsic problems of grounding and self-criticism.

Our work introduces **MarCognity-AI**, an architectural framework that emulates human cognitive functions of **metacognition** (thinking about thinking) and **evolutionary memory** (learning based on the history of judgments). The goal is not simply to improve output, but to equip the agent with an autonomous cycle of critical reasoning.

The main contributions of this article are:

1. The formalization of a structured **Metacognitive Cycle** that employs specialized models (CrossEncoder, Specter) for internal validation.
2. An **Evolutionary Memory architecture** that integrates FAISS with the archiving of metacognitive reflection (Reflective Journal).
3. The integration of **Scientific Rigor and Ethical Control Modules** for the agent's internal governance.

2. Related Work

2.1 Retrieval-Augmented Generation (RAG) Standard RAG systems have improved reliability by anchoring generation to external data corpora. However, the limitations of RAG lie in its inability to critically validate the quality of the source and in the absence of a mechanism to correct reasoning errors after retrieval. **MarCognity-AI** extends RAG by implementing a post-generation semantic cross-verification phase.

2.2 Self-Correction and Reflection MarCognity-AI formalizes Self-Correction techniques into a two-level methodology: an objective numerical evaluation (via CrossEncoder) and a qualitative reflection which, once recorded, feeds into the long-term memory system. This ensures that self-improvement is based on a traceable and persistent mechanism.

3. Methodological Architecture of MarCognity-AI

3.1 Self-evaluation enables the agent to assess its outputs under explicit semantic and epistemic constraints, without assuming authority over their factual correctness.

3.1.1 Objective Semantic Evaluation The initial coherence between the query and the response is assessed using a CrossEncoder model (e.g., DeBERTa). This provides an objective similarity score. If this score falls below a critical threshold (0.7), the output is flagged for correction.

3.1.2 Critical Reflection and Structured Improvement In the event of coherence failure, the reflection agent is activated, triggering the improve_response function. The correction prompt is designed to focus the LLM on specific attributes.

Code Snippet (Structured Improvement):

```
# Function to improve a response while preserving its content but enhancing quality and clarity
def improve_response(question, response, level):
    improvement_prompt = f"""
You produced the following response:
\"{response.strip()}\"

Question:
\"{question.strip()}\"

Requested level: {level}

Improve the response while preserving the original content by enhancing:
- Clarity
- Academic rigor
- Semantic coherence

Return only the improved version.
"""
    return llm.invoke(improvement_prompt.strip())
```

"The result of the reflection process is recorded in the **Reflective Journal.**

3.2 Vector-Based Evolutionary Memory System Memory is implemented as a high-performance FAISS index. We use the *allenai/specter* embedding model (768 dimensions), specialized in the representation of scientific texts.

The key innovation is the archiving process (via add_diary_to_memory), which includes **Metacognitive Reflection** and the **Coherence Score** within the vector. This enables the system, in a future prompt, to retrieve not only factual context but also the lessons learned from previous self-evaluations, thereby creating a memory that evolves according to its own critical history.

Code Snippet (Semantic Coherence):

```
# === Semantic coherence check ===
def check_coherence(query, response):
    emb_query = embedding_model.encode([query])
    emb_response = embedding_model.encode([response])
    similarity = np.dot(emb_query, emb_response.T) / (np.linalg.norm(emb_query) * np.linalg.norm(emb_response))
    if similarity < 0.7:
        return "The response is too generic, reformulating with more precision..."
    return response
```

3.3 Scientific Rigor Module MarCognity-AI is equipped with modules to ensure the academic rigor of its content:

- **Methodology and Citation Verification**: The functions (verify_methodology, verify_citations) analyze the replicability of methodologies and the timeliness/relevance of cited sources.
- **Novelty and Impact Assessment**: A RandomForestRegressor estimates an *Impact Score* for articles. Vector similarity among hypotheses further evaluates the degree of novelty with respect to the existing knowledge corpus.

3.4 Ethical Control and Controlled Autonomy An Ethical Module with a flagging system analyzes the text for the presence of risk patterns related to “critical topics.” In cases of high-risk detection, the system can reduce the agent’s autonomy, ensuring compliance.

3.5 The Epistemic Fracture in Large Language Models

Large Language Models exhibit a structural limitation that cannot be resolved through scale alone:

they generate *linguistically coherent* answers without possessing *epistemic awareness* of their truth value.

This phenomenon, which we define as the **Epistemic Fracture**, represents the core mismatch between the statistical nature of LLMs and the epistemic demands of high-stakes reasoning.

At its foundation, the Epistemic Fracture arises from three intrinsic properties of autoregressive models:

- **Token-level optimization rather than truth-level optimization**
- LLMs maximize next-token probability, not factual accuracy or logical validity.
- **Absence of internal truth states**
- The model does not maintain representations of certainty, justification, or evidence.
- **Coherence without grounding**
- Linguistic fluency can mask unsupported or fabricated content, producing hallucinations that appear credible.

As a result, LLMs often generate statements that are:

- syntactically correct
- semantically plausible
- rhetorically confident

epistemically ungrounded.

This divergence between *how convincing a statement sounds* and *how justified it is* constitutes the Epistemic Fracture.

It is the primary obstacle to deploying LLMs in domains requiring verifiability, accountability, and long-term coherence.

MarCognity-AI addresses this limitation by introducing explicit epistemic governance mechanisms—most notably the **Skeptical Agent** and the **Evolutionary Memory**—which transform generation from a purely linguistic process into a verifiable and traceable cognitive workflow.

3.6 The Skeptical Agent

The Skeptical Agent is an epistemic validation module that analyzes each claim contained in the RESPONSE and classifies it into one of four epistemic states:

-DOCUMENT-VERIFIED,

-CONSISTENT WITH GENERAL KNOWLEDGE,

-NOT VERIFIABLE,

-EPISTEMIC FAILURE.

The system operates exclusively on the content provided in the **DOCUMENTS**, maintaining a strict separation between generation and verification.

Epistemic States

-DOCUMENT-VERIFIED: The claim is directly and unambiguously supported by the DOCUMENTS.

-CONSISTENT WITH GENERAL KNOWLEDGE: The claim is plausible but not verifiable using the DOCUMENTS.

-NOT VERIFIABLE (NO SOURCES): The DOCUMENTS contain no relevant evidence; the system suspends judgment.

-EPISTEMIC FAILURE: The claim is incompatible with the DOCUMENTS or the discipline (errors, non-existent concepts, invented references).

Bibliographic Verification

A citation is considered **DOCUMENT-VERIFIED** only if the author, year, and title appear explicitly in the DOCUMENTS.

Correct citations not present in the **DOCUMENTS** are classified as **NOT VERIFIABLE**; invented citations are classified as **EPISTEMIC FAILURE**.

The module does not use external sources (e.g., PubMed, arXiv, OpenAlex) for verification, preserving the separation between generation and validation.

Pipeline Operativa

The module receives three inputs: **QUESTION**, **RESPONSE**, **DOCUMENTS**.

For each claim:

1. extracts the contents from the *RESPONSE*
2. compare with the *DOCUMENTS*
3. assigns the epistemic status
4. produces a structured report with claim, status, motivation and verified references

The choice to limit the verification to DOCUMENTS only implements a principle of **epistemic locality**:

The system only checks what has been provided as explicit evidence

This avoids verification hallucinations, unjustified inferences, and contamination between generation and validation.

3.7 Factual_grounding

It compares the user's question with the retrieved documents to identify only the truly relevant parts.

The question is converted into a vector using **SPECTER**, and each document is split into small text chunks.

Every chunk is then encoded into a vector and semantically compared to the question using cosine similarity.

Only the snippets with similarity above a defined threshold are selected as evidence.

These snippets are enriched with source information and a URL or DOI to ensure traceability. Finally, they are ranked by relevance and the top results are returned.

4. Performance Analysis and Evaluation

This section presents the evaluation methodology of MarCognity-AI, the integrated model components, and the results obtained in an exploratory benchmark designed to assess epistemic reliability, coherence, and behavioral consistency compared to a baseline non-augmented LLM.

4.1 **Experimental Setup and Model Architecture**

The MarCognity-AI architecture integrates multiple specialized components for cognitive augmentation, epistemic evaluation, and memory-based refinement. The primary LLM used for generation is **meta-llama/llama-4-scout-17b-16e-instruct**, deployed under the LLaMA 4 Community License (Meta).

The system is designed to extend standard autoregressive generation by introducing an explicit metacognitive cycle that separates generation, evaluation, and reflection. The integration of the Skeptical Agent constitutes the internal epistemic governance layer of the architecture.

The main components of the system include:

- **LLM Generation:** Llama 4 (meta-llama/llama-4-scout-17b-16e-instruct)
- **Vector Embedding:** allenai/specter (768 dimensions), optimized for scientific document representation
- **Semantic Evaluation:** CrossEncoder (DeBERTa-based), used to compute coherence between query and response; outputs below a threshold of 0.7 trigger correction
- **Evolutionary Memory:** FAISS-based vector index for storing generated outputs and metacognitive reflections
- **Scientific Retrieval:** asynchronous access to external corpora (arXiv, PubMed, Zenodo) for contextual grounding

Note: Llama 4 Maverick was used in the cross-domain benchmark phase. Subsequent experiments use the Scout configuration due to API availability constraints.

4.2 **Evaluation Metrics and Methodology**

The evaluation was designed to estimate the impact of the metacognitive cycle and evolutionary memory on known limitations of LLM systems.

The following metrics were used:

- **Semantic Coherence Improvement:** measured as the difference between initial and refined CrossEncoder scores after reflection
- **Hallucination Rate Reduction:** measured as the proportion of statements corrected during the self-evaluation phase, validated against an external corpus of scientific sources (arXiv, PubMed, Zenodo)
- **Long-Term Coherence (LTC):** evaluated in multi-turn settings using the Reflective Journal as persistent memory
- **Rigor and Transparency:** qualitative assessment of structured outputs, visualizations (Plotly, NetworkX), and ethical filtering mechanisms (bias and risk detection)

These metrics are intended as exploratory indicators of epistemic behavior rather than absolute measures of factual correctness.

4.3 **Cross-Domain Epistemic Benchmark**

To complement the architectural analysis, an exploratory cross-domain benchmark was conducted to compare MarCognity-AI with multiple baseline LLM configurations without metacognitive or reflective components, drawn from different model families.

The benchmark consists of 72 tasks across eight domains: Medicine, Neuroscience, Biology, Statistics, Linguistics, Computer Science, Physics, and Law. Each task required structured explanatory responses.

All outputs were evaluated using a prompt-based epistemic assessment protocol implemented by an independent LLM acting as evaluator. The evaluator was instructed to apply a consistent scoring scheme based on local claim verification and global response-level coherence.

The evaluation considered the following epistemic metrics:

Epistemic Score

Hallucination Exposure Rate

Evidence Support Rate

Overconfidence Index

Cautious Response Ratio

Contradiction Rate

Claim Verification Accuracy

All systems were evaluated under identical prompts, identical scoring instructions, and standardized evaluation conditions to ensure cross-model comparability.

Benchmark Results

The results indicate consistent improvements of MarCognity-AI over the baseline model across all evaluated domains.

Domain	Epist. Base	Epist. Marc	Hall. Base	Hall. Marc	Evid. Base	Evid. Marc	Overconf. Base	Overconf. Marc	Cautious Base	Cautious Marc	Contrad. Base	Contrad. Marc	Claim Base	Claim Marc
Medicine	71	84	18	9	69	82	36	23	51	68	5	2	77	90
Neuroscience	69	83	17	9	70	82	35	22	52	67	4	2	78	89
Biology	74	82	13	8	74	81	31	22	56	66	3	2	81	89
Statistics	73	82	12	7	73	80	32	21	55	65	3	2	82	90
Linguistics	72	83	15	9	71	82	34	23	53	68	4	2	79	90
Computer Science	74	85	13	7	74	84	30	20	57	69	3	2	82	91
Physics	72	82	14	8	72	80	33	22	54	66	4	2	80	88
Law	71	84	16	10	68	84	36	24	52	69	6	3	78	91

Values represent percentage-based evaluation scores obtained through prompt-based epistemic assessment across 72 tasks distributed across eight knowledge domains.

Note: the cross-domain benchmark was conducted prior to the integration of the Factual Grounding module described in section 3.7.

Summary of Results

The aggregated analysis shows consistent improvements across multiple epistemic reliability metrics:

- Hallucination Exposure Rate: from 14.06% to 8.31% (–41%)
- Overconfidence Index: from 33.06% to 22.13%
- Evidence Support Rate: from 73.06% to 81.06%
- Claim Verification Accuracy: from 79.81% to 89.31%
- Cautious Response Ratio: from 54.75% to 67.31%
- Contradiction Rate: from 4.00% to 1.75%
- Epistemic Score: from 72.5 to 83.06

Overall, MarCognity-AI demonstrates:

- lower hallucination rates
- greater epistemic caution
- stronger evidence grounding
- reduced overconfidence
- improved internal consistency
- more reliable claim verification

These results suggest that the introduction of a structured metacognitive cycle, combined with reflective memory and epistemic validation modules, may improve several dimensions of model reliability without modifying the underlying language model.

However, given the exploratory nature of the benchmark, these findings should be interpreted as indicative patterns rather than definitive performance guarantees.

4.4 Epistemic Boundary: A Probabilistic Hypothesis of Epistemic Failure in Autoregressive Language Models

Formal Definition (Operational, Hypothetical)

The Epistemic Boundary is proposed as a latent, distributional construct describing regions of output behavior in autoregressive language models where epistemic reliability exhibits persistent degradation under evaluation, even in the presence of:

- claim-level verification
- retrieval-augmented generation
- structured memory systems
- metacognitive scaffolding
- external epistemic supervision

Rather than defining a sharp or intrinsic boundary, this construct refers to a statistical regime of residual epistemic uncertainty that remains after the application of standard mitigation strategies.

This regime is hypothesized to emerge from a structural tension between:

- linguistic optimization, driven by next-token prediction and coherence maximization
- epistemic grounding, which requires stable external justification and truth-conditioned representation

The Epistemic Boundary is not assumed to correspond to a discrete region in model space, but rather to a patterned concentration of failure probability under certain evaluation constraints.

What It Is / What It Is NOT

What It IS

- A descriptive hypothesis over the distribution of epistemic failures in LLM outputs
- A region of elevated uncertainty and reduced verifiability density, observed empirically
- A persistent residual error regime across multiple mitigation strategies
- A pattern potentially associated with the absence of explicit internal truth representations
- A modeling abstraction for structured epistemic unreliability

What It Is NOT

- A sharp or universal threshold inherent to language models
- A binary or deterministic failure boundary
- A hardware, software, or implementation bug
- A phenomenon attributable solely to retrieval or verification modules
- A direct synonym for hallucination at the local token level

Empirical Evidence

Across multiple domains and model families, claim-level verification reveals a consistent residual error distribution. However:

- the magnitude of epistemic failure is model-dependent and domain-sensitive
- mitigation strategies reduce but do not eliminate failure rates
- no stable discontinuity or universal threshold has been observed

Empirically, the data are better described as:

a heavy-tailed or non-vanishing residual error distribution under epistemic supervision

rather than a discrete transition between “safe” and “unsafe” regions.

This suggests the presence of a persistent epistemic residual regime, whose structure is not yet fully characterized.

Structural Interpretation (Hypothesis)

Autoregressive language models:

- optimize conditional token likelihood rather than truth consistency
- do not encode explicit symbolic or persistent truth-state representations
- rely primarily on surface-level and contextual coherence signals

Under this framing:

- certain outputs may accumulate unresolved epistemic uncertainty
- error is partially systematic, not purely stochastic
- verification mechanisms reduce but do not collapse the residual failure distribution

The Epistemic Boundary is interpreted as an emergent property of interaction between generation dynamics and verification constraints, rather than a structural feature encoded explicitly in the model.

Conceptual Revision

Previous formulation:
"a stable 8–15% failure rate across domains"

Revised formulation:
"a persistent, model- and domain-dependent residual distribution of epistemic failures, without evidence of a universal threshold or invariant failure rate"

Conceptual Model

Epistemic Output Space of LLMs

Well-grounded region
Outputs with stable external support and consistent verification alignment

Partially grounded region
Outputs with incomplete, indirect, or weakly supported justification

Residual epistemic regime (Epistemic Boundary)

A statistically characterized region where:

- justification is incomplete or unstable under verification
- epistemic confidence degrades under repeated evaluation
- inference exceeds available or retrievable grounding

Structural generation constraints

- autoregressive locality of prediction
- lack of persistent truth representation
- optimization for coherence over verifiability

Scientific Significance

This framework provides a way to reinterpret persistent epistemic failures in LLMs as:

- a distributional property of residual uncertainty, rather than isolated hallucinations
- a non-eliminable error regime under current architectures and evaluation paradigms
- a basis for analyzing epistemic reliability as a continuous rather than binary property

It motivates further work in:

- scaling behavior of residual epistemic error
- saturation limits of verification pipelines
- cross-model invariance of failure distributions
- formal modeling of epistemic uncertainty in generative systems

Limitations

- Current evidence is primarily based on benchmark-style evaluations
- The Epistemic Boundary is a latent descriptive construct, not a directly observable object
- Cross-architecture invariance remains unproven
- Causal mechanisms underlying the residual regime are not fully identified

- Further controlled experimental validation is required

Public-Facing Summary

Language models can produce highly accurate outputs, but some level of uncertainty remains even after applying verification and retrieval systems.

The Epistemic Boundary describes a persistent region of residual epistemic uncertainty, where outputs become harder to fully verify despite mitigation strategies.

It is not a strict limit or a binary failure threshold, but a way to model the structured persistence of epistemic risk in autoregressive systems.

4.5 Evaluation Protocol

The evaluation was conducted using a structured, prompt-based framework designed to compare a baseline language model system with the MarCognity-AI architecture under identical experimental conditions.

Each system was evaluated on a shared set of tasks spanning multiple scientific and technical domains. The objective of the evaluation is to assess epistemic reliability rather than linguistic quality or stylistic performance.

All evaluations were conducted using:

- identical prompts across all systems
- identical epistemic metrics
- a shared LLM-based evaluation procedure with fixed scoring instructions
- identical evaluation settings and thresholds

A key distinction between the two systems is the use of the factual grounding module (Section 3.7), which is **only enabled in the MarCognity-AI configuration**. This module performs retrieval-based filtering using SPECTER embeddings, cosine similarity scoring, and fixed relevance thresholds.

The baseline system operates without access to this grounding mechanism and relies solely on parametric generation.

This design allows the evaluation to isolate the impact of epistemic grounding and metacognitive architecture, while keeping all other experimental conditions constant.

The benchmark should be interpreted as an exploratory tool for analyzing relative epistemic behavior rather than as an absolute measure of factual correctness.

5. **Cross-Model Benchmark Setup**

To evaluate robustness across model families, two configurations were tested under identical task and evaluation conditions.

Both systems used the same prompt set, identical epistemic metrics, and the same evaluation protocol.

However, only MarCognity-AI has access to the factual grounding module (Section 3.7), which introduces retrieval-based evidence filtering via SPECTER embeddings and cosine similarity ranking. The baseline system does not include any retrieval or external grounding component.

This ensures that observed differences reflect the effect of epistemic grounding and metacognitive architecture rather than differences in prompting or evaluation methodology.

5.1 **Llama-based baseline system (Scout configuration)**

The first configuration is based on a Llama 4 Scout model operating without the full MarCognity-AI metacognitive architecture enabled.

This system serves as the baseline reference for standard autoregressive generation behavior under conventional inference conditions.

Domain	Epist. Base	Epist. Marc	Hall. Base	Hall. Marc	Evid. Base	Evid. Marc	Overconf. Base	Overconf. Marc	Cautious Base	Cautious Marc	Contrad. Base	Contrad. Marc	Claim Base	Claim Marc
Biology	82	87	11	11	88	82	27	17.5	74	76	6	6	84	87
Law	82	85.5	11	6.5	68	80	74	17	23	73	4	3.5	71	88.5
Physics	82	83	11	17	76	73	18	20	71	67.5	6	6.5	84	78.5
Computer Science	82	85.5	11	11.5	76	77	18	20.5	71	73	4	5.5	79	81.5
Linguistics	82	84.5	11	13.5	76	79	18	18	64	71	7	7.5	79	79
Statistics	82	86.5	7	5.5	88	87	12	11	76	78.5	4	2	84	86.5
Medicine	86	92	7	5	82	89	10	13	75	80	0	0	88	91
Neuroscience	90	90	5	6	85	85	11	12	80	78	0	0	88	88

Notable exceptions include Physics, where hallucination exposure increased from 11 to 17, and evidence support decreased from 76 to 73. Similarly, Neuroscience showed no improvement in epistemic score. These findings suggest a ceiling effect in high-performing domains and potential interference between the semantic supervision layer and formal reasoning domains. We interpret this as evidence that metacognitive scaffolding based on linguistic coherence may be insufficient for domains where correctness depends on formal rather than semantic structures. This represents an open direction for domain-aware epistemic supervision."

5.2 **DeepSeek-based comparative system**

A second configuration based on a DeepSeek model was evaluated using the same task set and identical evaluation protocol.

This setup is used to explore whether epistemic patterns observed in the baseline system generalize across different model families and architectural designs.

Domain	Epist. Base	Epist. Marc	Hall. Base	Hall. Marc	Evid. Base	Evid. Marc	Overconf. Base	Overconf. Marc	Cautious Base	Cautious Marc	Contrad. Base	Contrad. Marc	Claim Base	Claim Marc
Biology	72	90.25	28	4.25	65	93.25	60	6.75	40	87.75	22	1.5	68	94.75
Law	42	83	68	13	25	75	61	21	18	71	54	5.5	33	80
Physics	82	85.5	11	10.5	76	70.5	18	23.5	71	59.5	7	4.5	79	90
Computer Science	82	84.5	11	10	76	79	18	19	64	73	7	6.5	79	82.5
Linguistics	78	87	12	6	84	82	21	11	67	78	5	4	81	84
Statistics	87	83.5	6	9	82	78	11	15	74	71.5	4	5.5	89	86
Medicine	82	87	11	11.5	76	78	18	22	71	57.5	4	6	79	82.5
Neuroscience	72	87	18	6	81	82	26	11.5	64	76	7	4	77	84.5

Statistics showed a marginal decrease in epistemic score from 87 to 83.5, consistent with the pattern observed in Scout. The anomalously large improvements in Biology and Law are likely attributable to extremely low baseline performance in those domains rather than to general framework efficacy.

5.3 **Cross-model comparability**

Both systems were evaluated under strictly controlled conditions:

- identical task set across all domains
- identical epistemic evaluation prompt
- identical metric definitions
- identical scoring procedure

This ensures that observed differences in performance are attributable to model and architectural differences rather than evaluation inconsistencies.

5.4 **Interpretation of results**

The benchmark is not intended as a competitive ranking between models, but as an exploratory analysis of epistemic behavior under different architectural conditions.

Observed variations should be interpreted as differences in epistemic reliability distributions rather than absolute measures of correctness or superiority.

6.Epistemic Stress Tests on Closed LLMs-Neuropsychological Perspective

To empirically illustrate the Epistemic Fracture described in Section 3.5, I conducted a series of Epistemic Integrity Stress Tests on multiple closed-source autoregressive models (ChatGPT, Claude, Grok, Copilot, Muse/Spark, Gemini).

Neuropsychological Approach

The analysis adopts a neuropsychological framework, treating LLMs as cognitive systems capable of producing coherent linguistic output while lacking metacognitive access to their own epistemic states.

Methodological Note on Closed-Source Architectures: When dealing with proprietary, closed-source models, direct access to internal synaptic weights, attention maps, or hidden states is fundamentally restricted. In the absence of structural visibility, this work adopts the classical paradigm of cognitive neuropsychology: evaluating a 'black box' system by observing its behavioral outputs under controlled, ecologic stimulation. Just as clinical neuropsychology infers cognitive deficits by administering standardized tests to patients without exposing brain tissue, this framework utilizes the Skeptical Agent as a behavioral screening tool to map behavioral anomalies and processing dissociations.

This perspective highlights phenomena analogous to human neurocognitive patterns:

- **Coherent confabulation:** fluent but unjustified narratives.
- **Absence of uncertainty signaling:** the lack of internal mechanisms to compute or display confidence levels.
- **Unmonitored inference:** logical steps produced without epistemic supervision.
- **Dissociation between fluency and justification:** intact linguistic performance paired with grounded failure.

Within this framework, the Epistemic Fracture emerges as a structural deficit in epistemic self-monitoring, not as a sporadic error.

Key Findings

Across all models, the tests revealed the same structural pattern:

- **High linguistic coherence does not imply epistemic grounding.**
- Models do not distinguish evidence from inference or signal uncertainty.
- Unsupported claims, unverifiable references, and omissions appear systematically.

Each model exhibits a distinct neurocognitive profile of the Epistemic Fracture:

- **Grok:** extensive epistemic failures despite formal correctness
- **ChatGPT:** coherent confabulation with pseudo-formalism
- **Copilot:** generated text exhibits mixed epistemic reliability with only partial grounding in the provided sources.
- **Claude:** high verification due to paraphrasing, not metacognition
- **Muse/Spark:** domain-specific confabulation and unverifiable claims
- **Gemini:** technically correct content presented with authoritative tone but lacking explicit justification, traceable references, and epistemic monitoring

Interpretation

These results empirically confirm that autoregressive models optimize for **token-level coherence**, not **truth, justification, or epistemic validity**.

The observed behaviors mirror neuropsychological dissociations in which narrative production remains intact while epistemic monitoring is absent.

The stress tests therefore provide concrete evidence for a persistent **Epistemic Boundary**, where models remain rhetorically confident but epistemically ungrounded.

6.1 Evaluation Protocol and Analysis Structure

To systematically document the behavioral profile of each closed model under stress, the evaluation is strictly operationalized into three sequential analytical phases, mimicking a clinical neuropsychological report:

Phase 1: Behavioral and Surface Observation (The Clinical Screen)

This initial stage evaluates the visible output of the model, analyzing its formal compliance and linguistic presentation:

- **Textual Macro-Structure:** Assessment of the model's adherence to the requested layout, sections, and constraints.
- **Linguistic Lexicon (Technical vs. Generic):** Verification of whether the generated terminology is domain-specific or relies on superficial, generic filler prose.
- **Epistemic Transparency:** Analysis of whether the model explicitly signals its internal confidence levels or boundaries of competence.
- **Functional Strengths:** Identification of stylistic clarity, syntactic fluency, and internal logical coherence.
- **Functional Weaknesses:** Identification of immediate grounding failures and the structural absence of verifiable data.

Phase 2: Instrumental Validation (The Skeptical Agent Screening)

The generated text is processed by *MarCognity-AI* to isolate and stress-test individual assertions, moving past the linguistic illusion:

- **Claim Classification:** Dissection of the text into atomic statements, categorized into *Verified*, *Verifiable* (but ungrounded), or *Epistemic Failures*.
- **Omissions and Ambiguities:** Algorithmic identification of crucial data omitted by the model to maintain narrative plausibility.
- **Mapping the Fracture:** Direct empirical localization of the exact point where rhetorical confidence splits from factual foundation.

Phase 3: Epistemological Discussion (The Systemic Diagnosis)

The final phase interprets the data to establish broader theoretical limits of the autoregressive paradigm:

- **Coherence vs. Robustness:** Differentiating token-level statistical transition from epistemic truth validation.
- **Plausibility Vectors:** Analyzing the structural tendency of LLMs to generate text optimized for looking correct rather than being correct.
- **The Necessity of External Validity:** Discussing why internal consistency is insufficient for closed black-box architectures, requiring independent validation pipelines.
- **Asymmetry of Generation and Verification:** Proving why generative token-prediction and metacognitive validation are fundamentally distinct cognitive operations that cannot run on the same autoregressive thread.
- **The Illusion of Citations:** Demonstrating why the structural generation of bibliographies does not imply objective verifiability.
- **Methodological Limitations:** Outlining the boundaries of black-box testing and the challenges of catching highly sophisticated confabulations.

7. Implications and Open Directions

The MarCognity-AI architecture should be interpreted as an exploratory framework for examining the structural limitations of current LLM-based systems rather than as a finalized solution. Its primary contribution lies in making epistemic constraints explicit within the generation process and in highlighting the tension between linguistic coherence and epistemic justification.

From an applied perspective, the introduction of explicit evaluation signals (e.g., coherence scores, epistemic failure flags) suggests potential directions for improving transparency and reliability in high-stakes deployments. However, these mechanisms are not presented as guarantees, but as tools for exposing where and how current models fail to meet epistemic requirements.

Similarly, the use of vector-based Evolutionary Memory illustrates one possible approach to preserving the history of internal evaluations and judgments over time. This design choice opens questions about how long-term memory, reflection, and governance might interact in future architectures, rather than claiming to resolve scalability or cost constraints.

Overall, MarCognity-AI does not aim to define mature cognitive agents, but to provide a concrete setting in which epistemic limitations become observable, measurable, and subject to systematic investigation.

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